

Lab-01

Name: Munazza Ishfaq Allies Shaista

Reg No: 2024-BSE-056

Course Code: Cloud Computing

Section: BSE-5B

GitHub Username: munazzaishfaqalliesshaista056

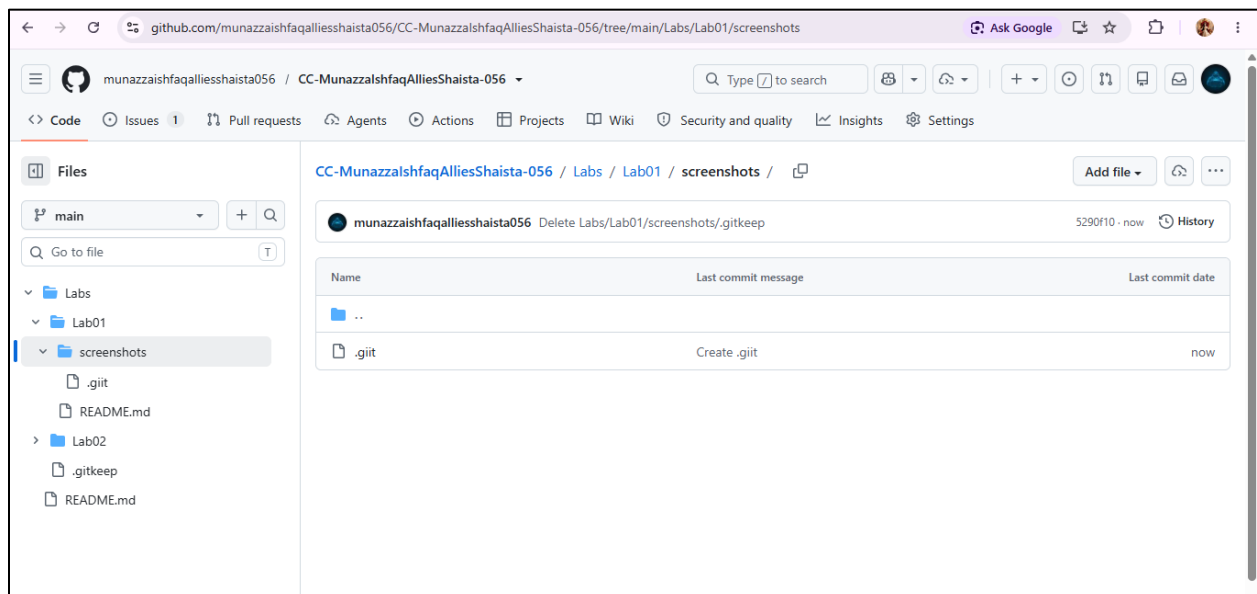
Lab Title: **GitHub Identity, Student Portal Registration, and Ubuntu Server Installation**

Getting Started

First, I established the required folder directory structure inside my CC repository to store all the lab proof screenshots for automatic grading.

- **Directory Structure Verification**

The required directory Labs/Lab01/screenshots/ was created successfully.



Created directory structure Labs/Lab01/screenshots/ in the CC repository.

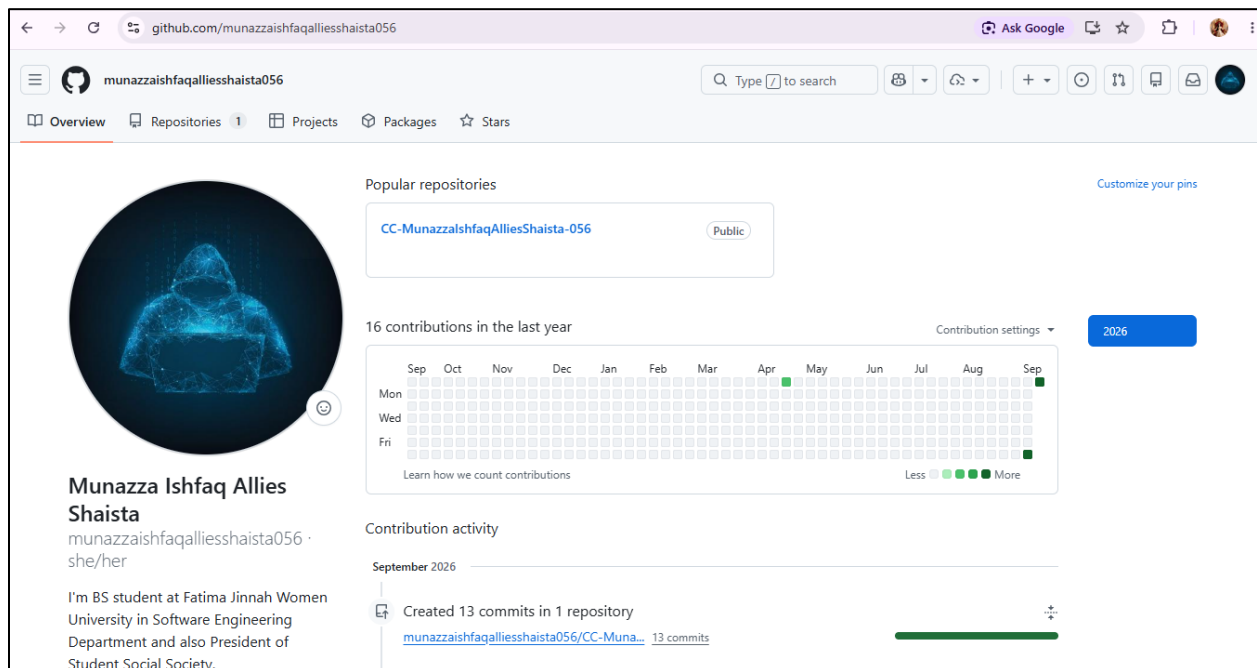
Task 1

Create and Configure Your GitHub Account

In this task, a professional GitHub account was set up ensuring identity consistency across the lab requirements.

- **Step 1: GitHub Account Setup**

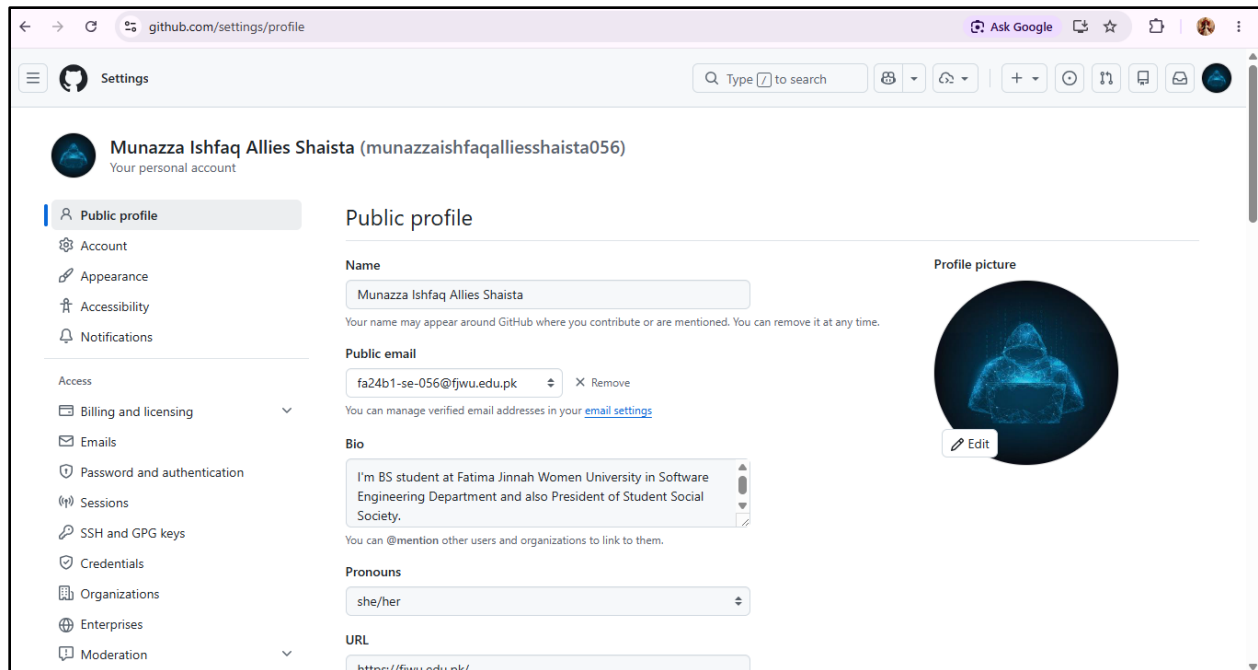
A personal GitHub account was created using a professional username (lowercase letters/numbers, no spaces) based on my actual name.



GitHub profile showing the chosen professional username.

- **Step 2: Public Profile Configuration**

Navigated to **GitHub Settings > Public Profile** and updated the **Name** field with my complete actual name.



GitHub public profile settings showing the completed Name field with my actual full name.

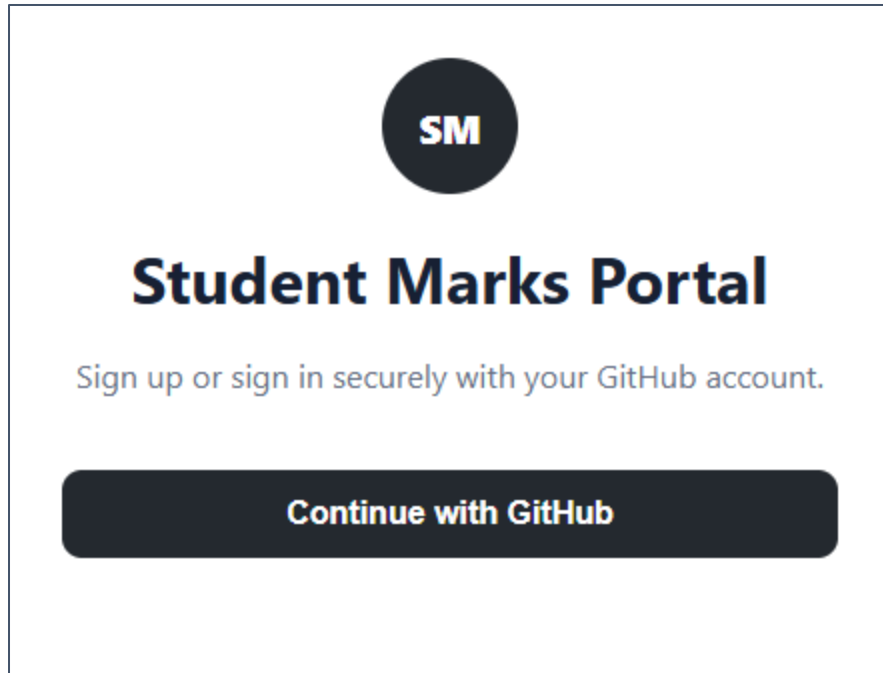
Task 2

Register in the Student Marks Portal

In this task, registration was completed on the Student Marks Portal using the GitHub account created in Task 1.

- **Step 1: Portal Access**

Navigated to the Student Marks Portal login interface.



Student Marks Portal landing and login page.

- **Step 2: Authentication and Profile Sync**

Logged into the portal using the **Continue with GitHub** option to link the newly configured GitHub identity.



Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-56

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section B

Submit enrollment

Sign out

GitHub profile information successfully linked and visible inside the Student Marks Portal.

- **Step 3: Student Enrollment Submission**

Entered the actual registration number, selected the appropriate course code, and specified the correct section before submitting the enrollment.

STUDENT MARKS PORTAL

Course enrollments

Sign out



Munazza Ishfaq Allies Shaista

2024-BSE-56

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section B

Waiting for administrator approval

Pending approval

Confirmation of submitted/pending enrollment with complete student details.

Task 3

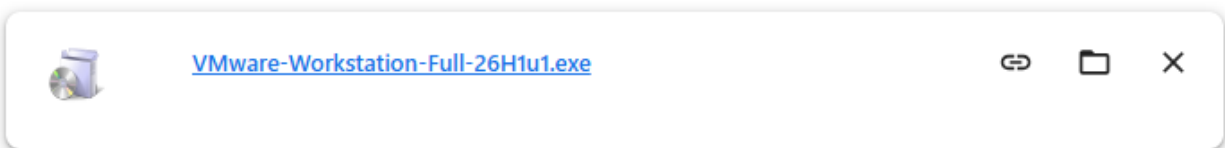
Download VMware Workstation Pro and Ubuntu Server

In this task, the required virtualization software was verified, the Ubuntu Server ISO image was downloaded, and disk space availability was confirmed.

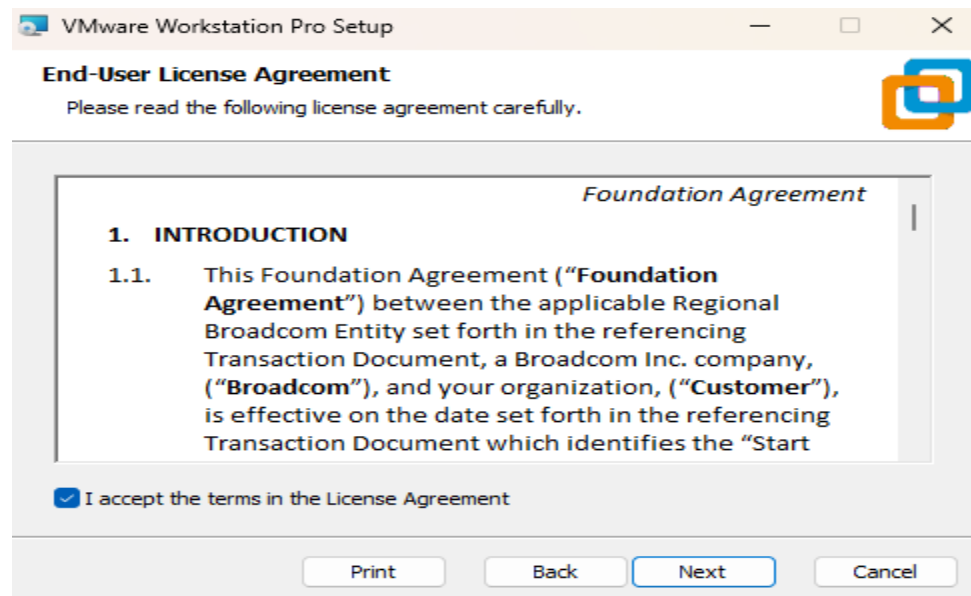
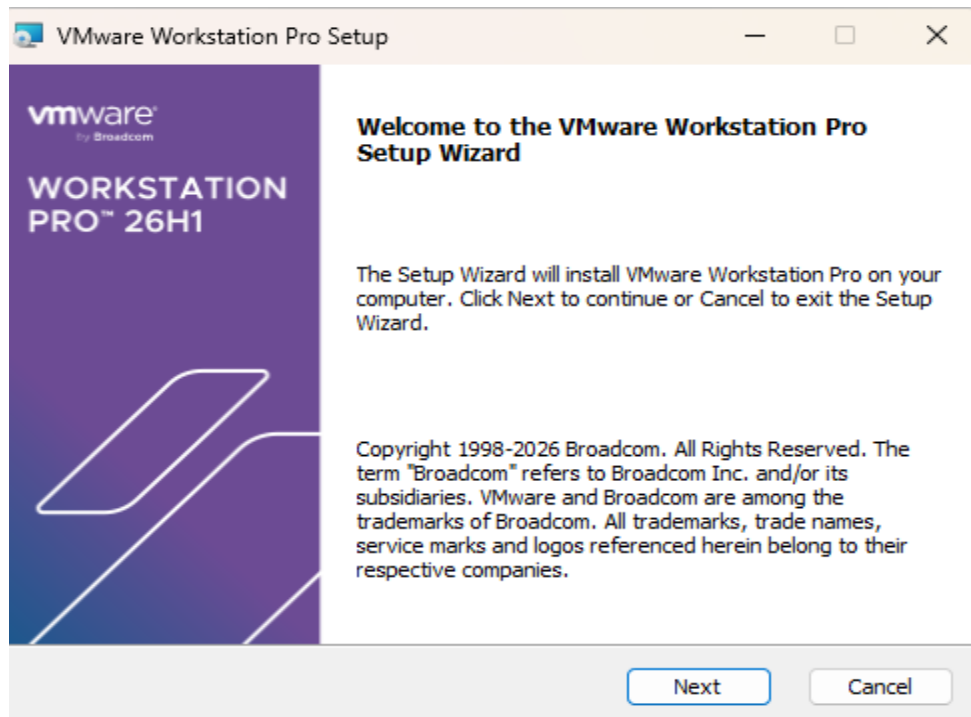
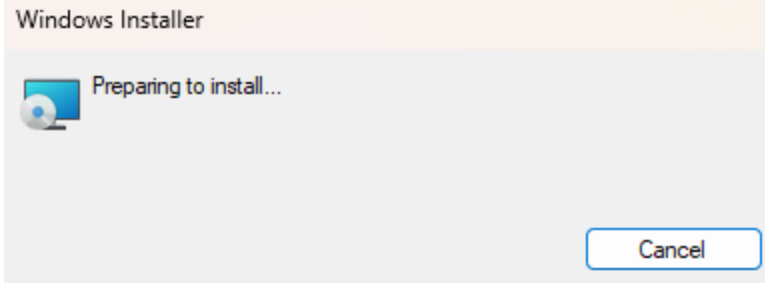
- **Step 1: Intall VMware Workstation Pro**

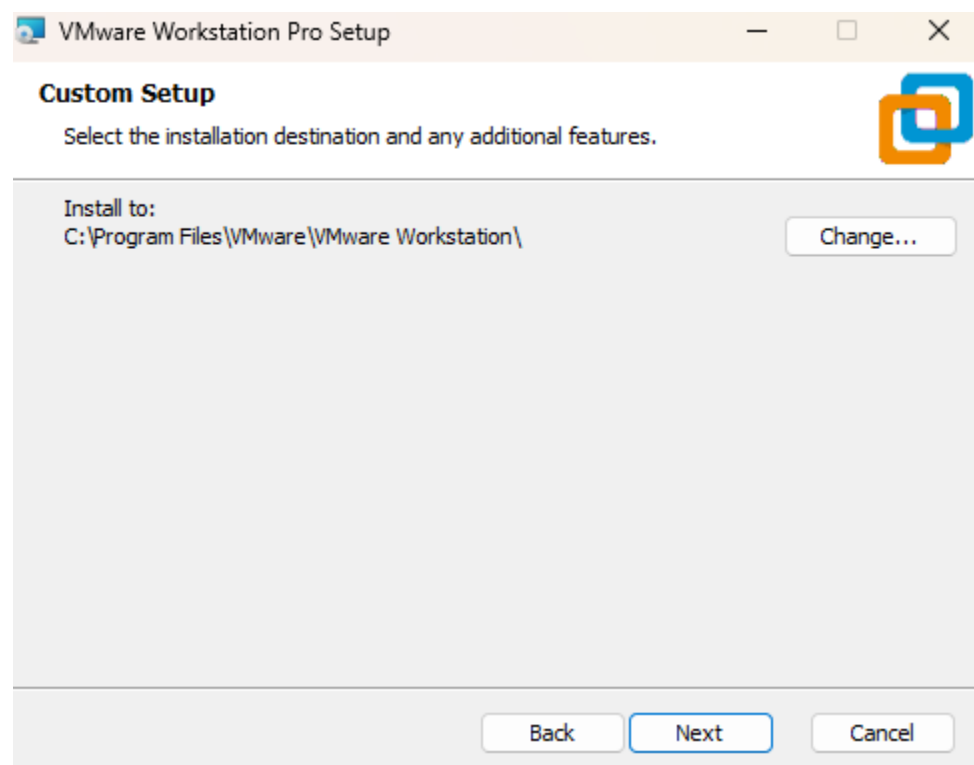
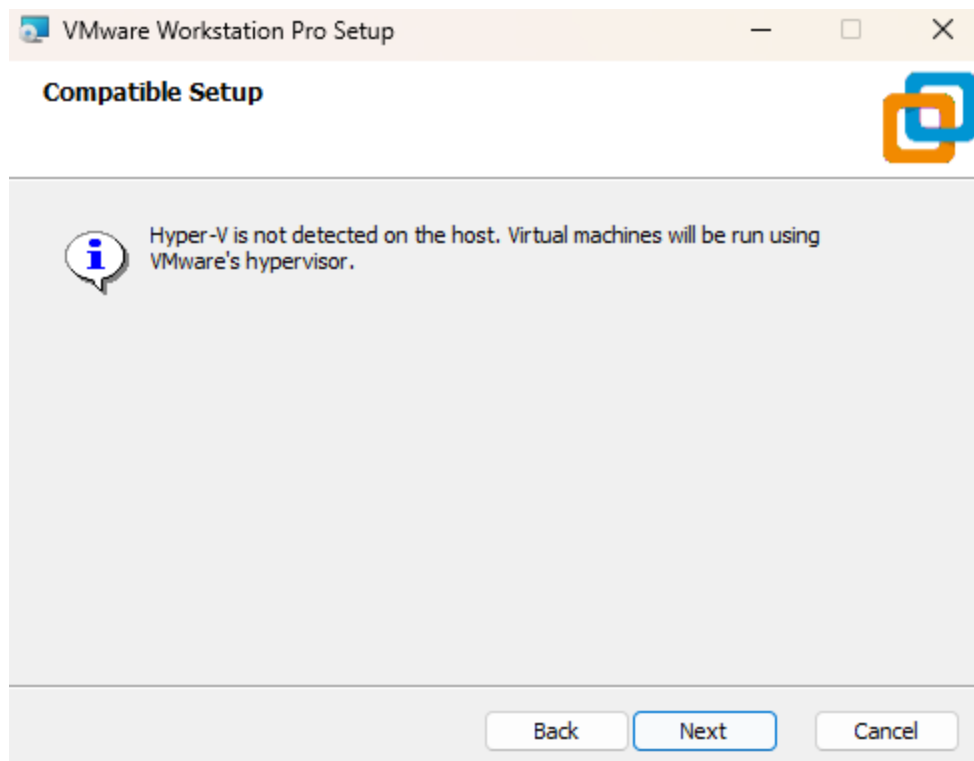
Download VMware workstation Pro from this Official website:

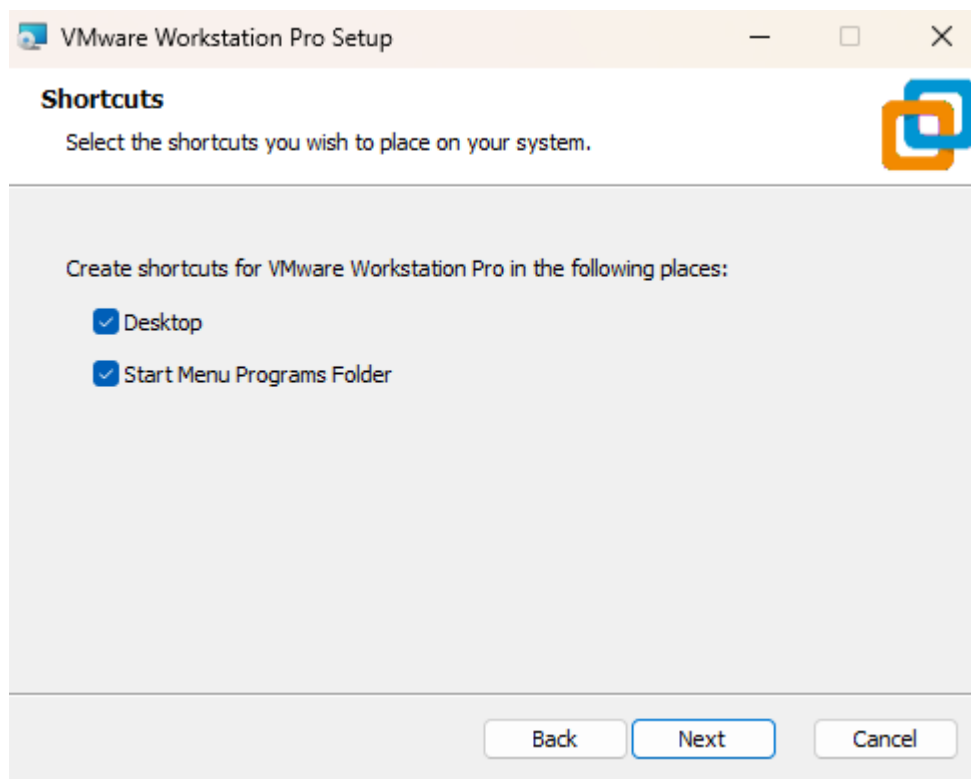
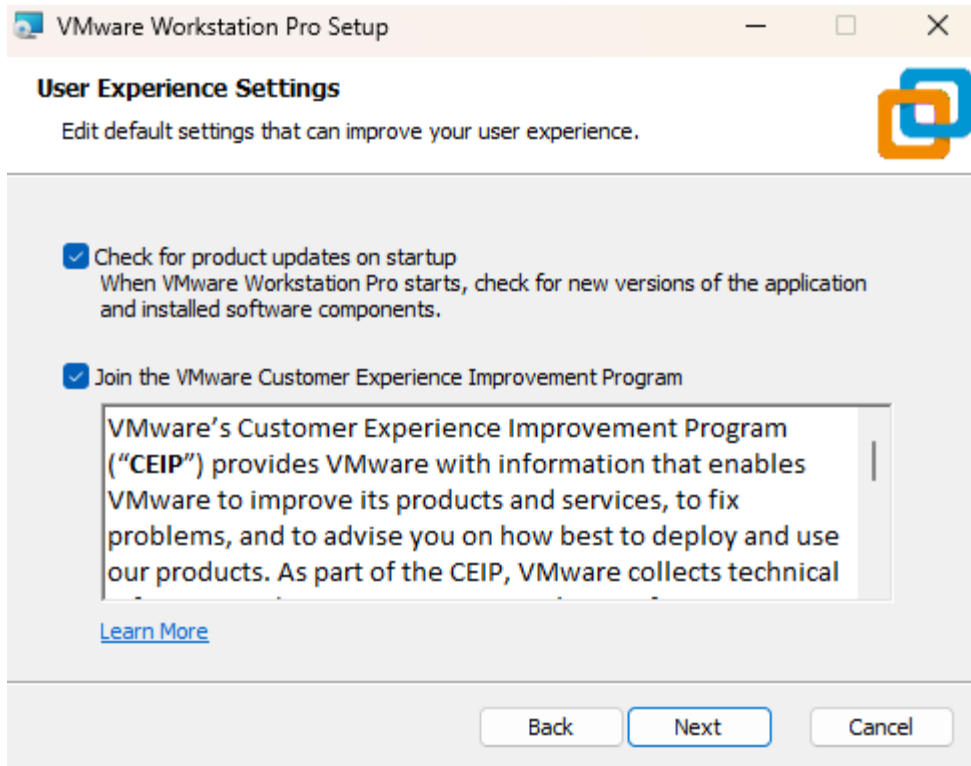
<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

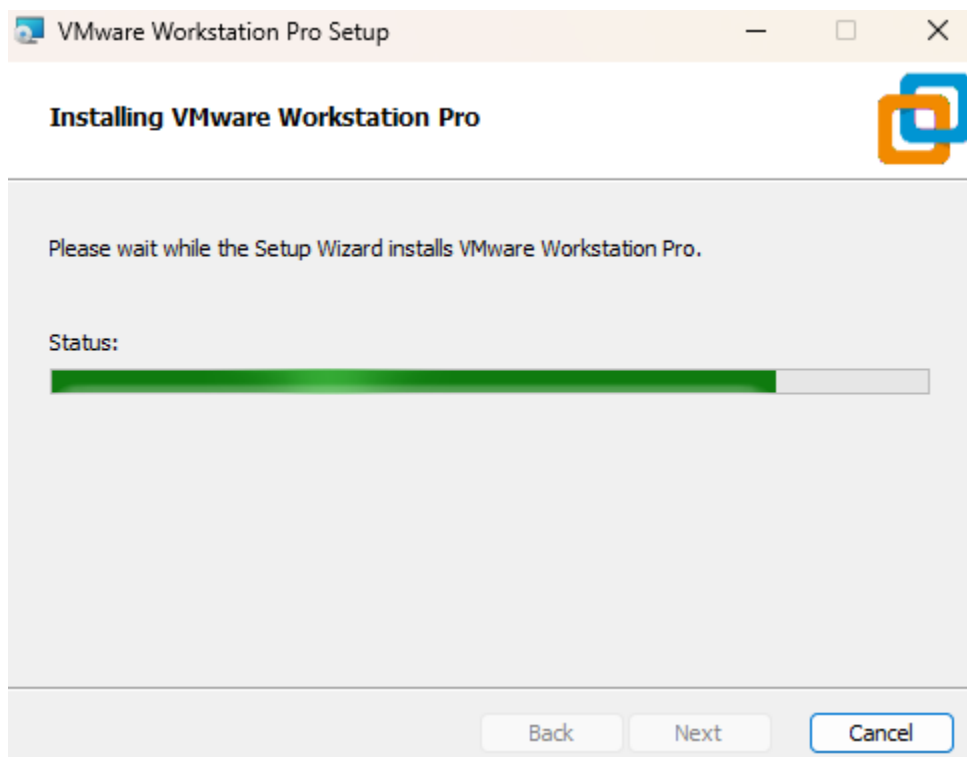
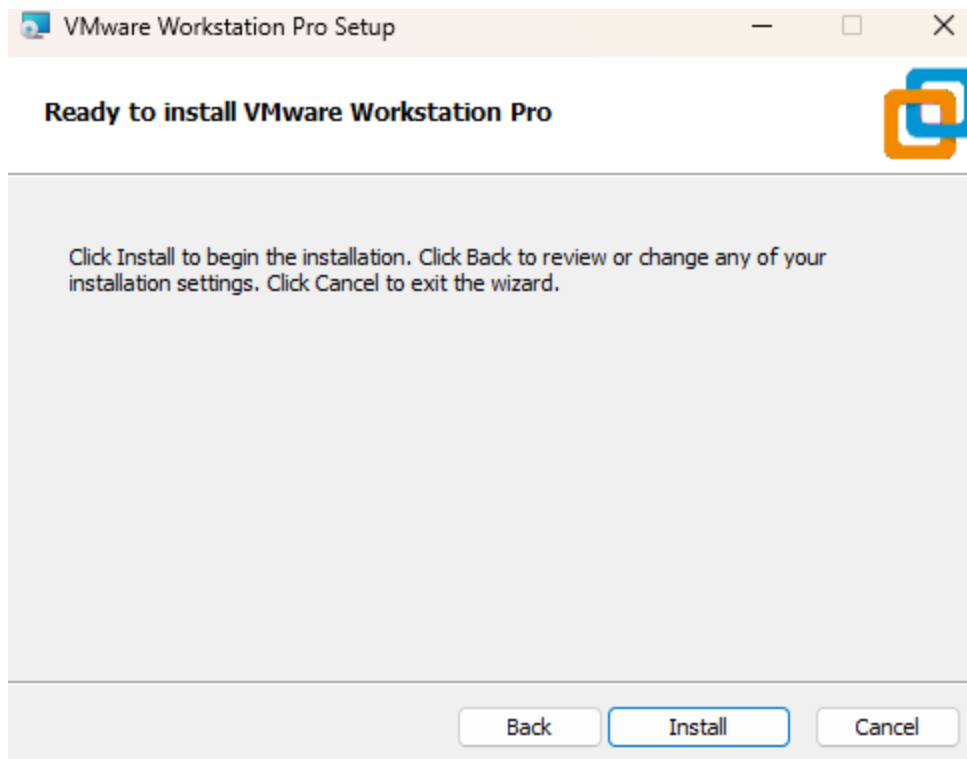


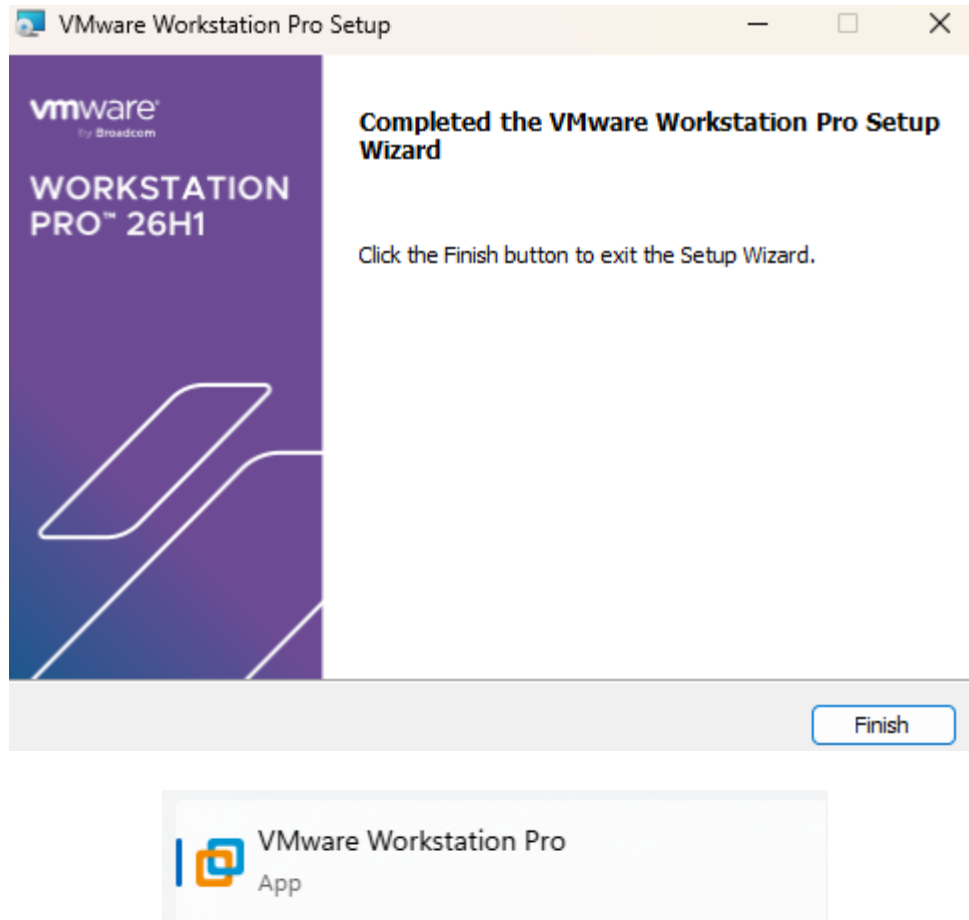
Run this file as an Administrator and Install VMware Workstation



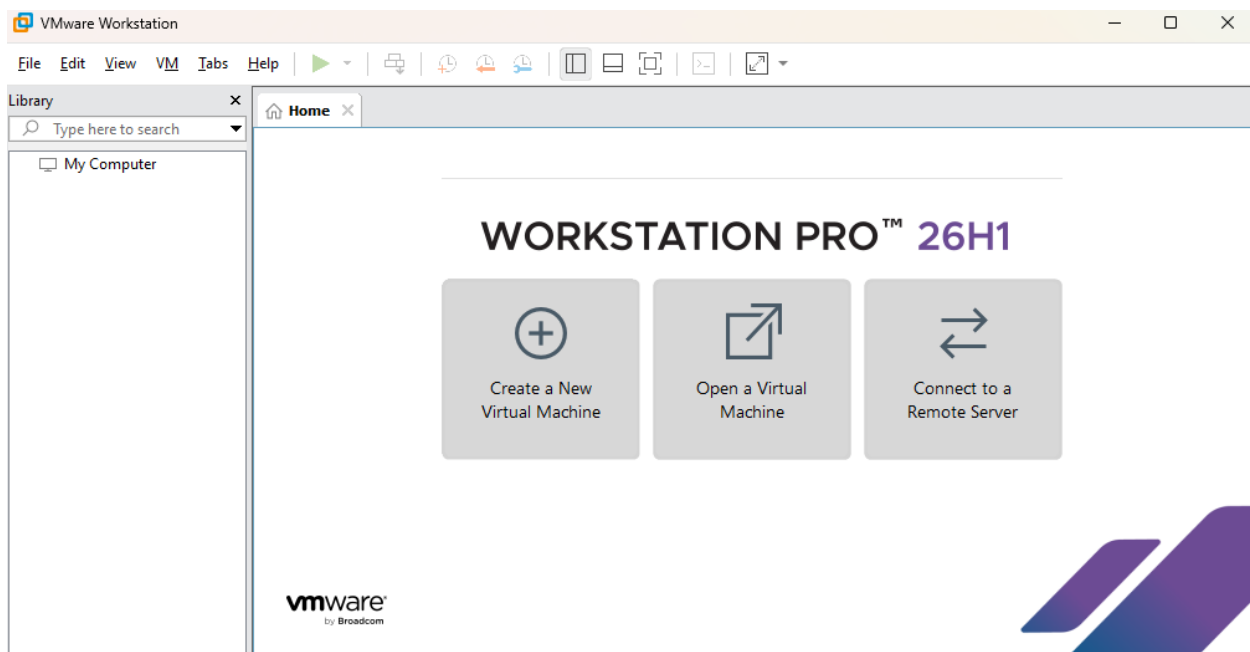








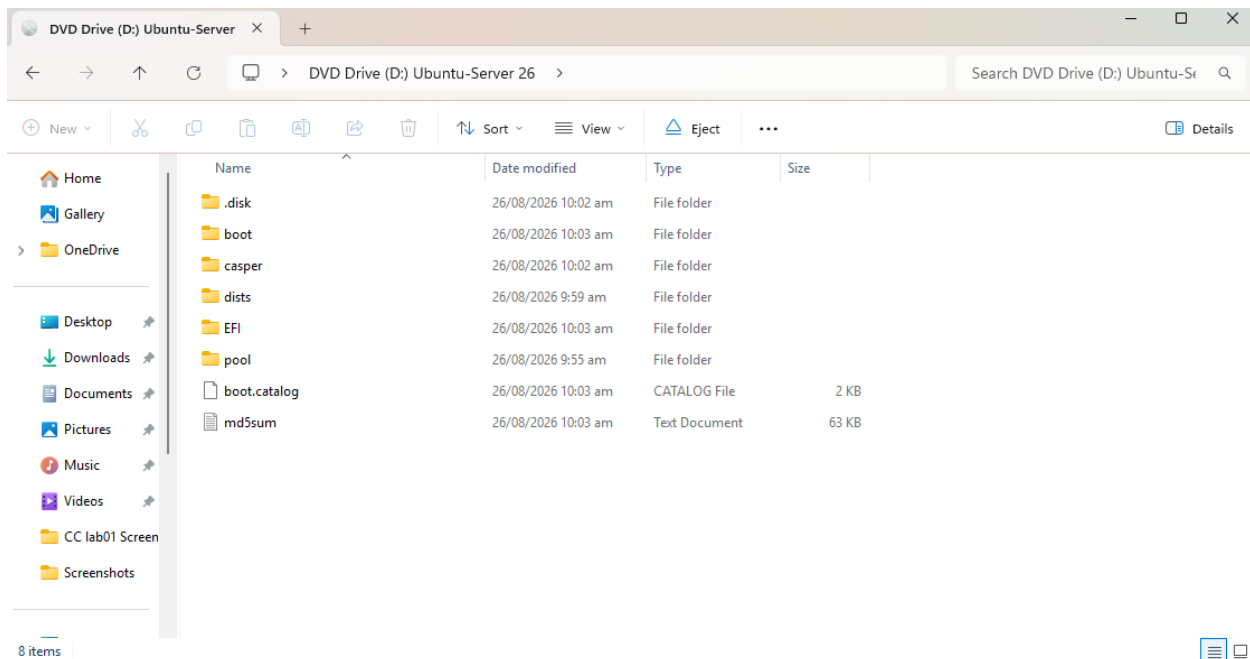
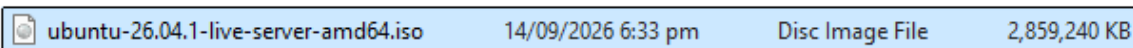
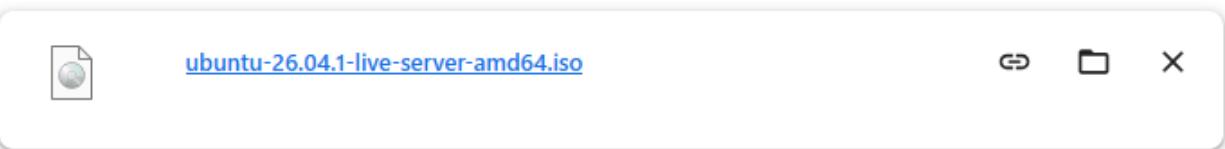
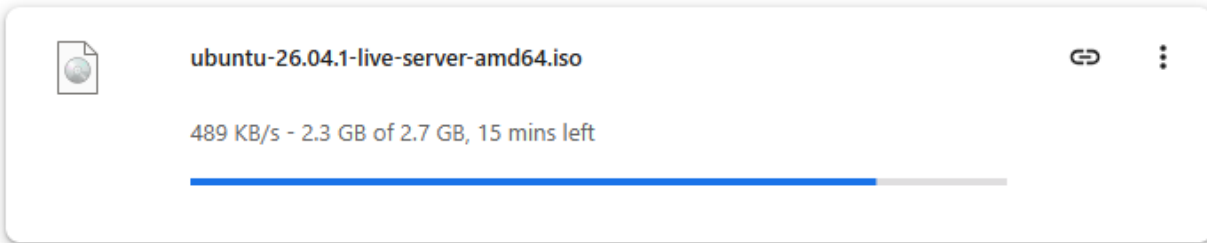
VMware workstation:



VMware Workstation Pro running successfully on the host operating system.

- **Step 2: Download Ubuntu Server LTS ISO**

Navigated to the official Ubuntu download page and downloaded the latest Ubuntu Server LTS ISO image file onto the local machine.



Downloaded Ubuntu Server LTS ISO file displayed in File Explorer with its full filename.

- **Step 3: Storage Space Verification**

Checked host drive properties to ensure more than 20 GB of free storage space is available for allocating the virtual machine disk.



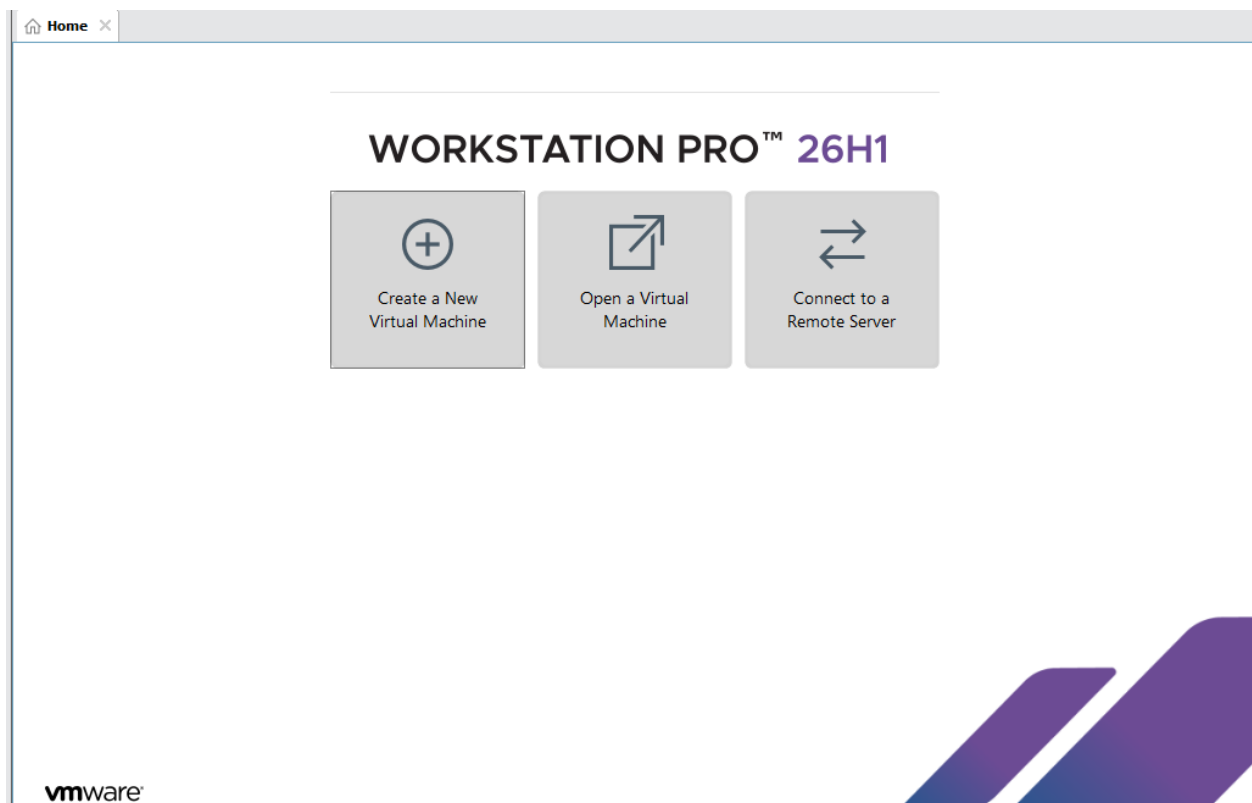
Host storage drive properties confirming more than 20 GB of available free disk space.

Task 4: Create the Ubuntu Server Virtual Machine

In this task, a new virtual machine was configured in VMware Workstation Pro for installing Ubuntu Server.

- **Step 1: Launch Virtual Machine Wizard**

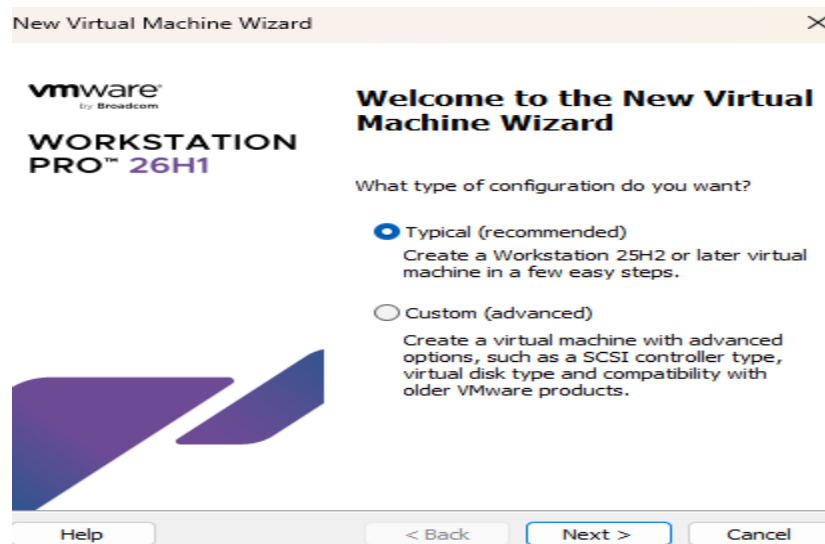
Opened VMware Workstation Pro and initiated the **Create a New Virtual Machine** wizard.



VMware New Virtual Machine Wizard launch screen.

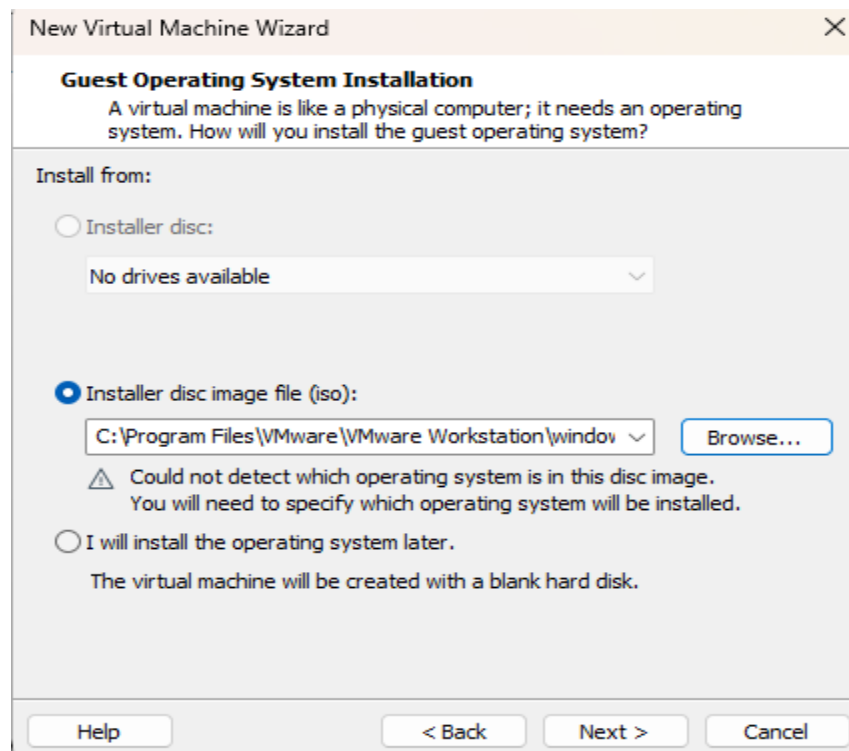
- **Step 2: Select Configuration Type**

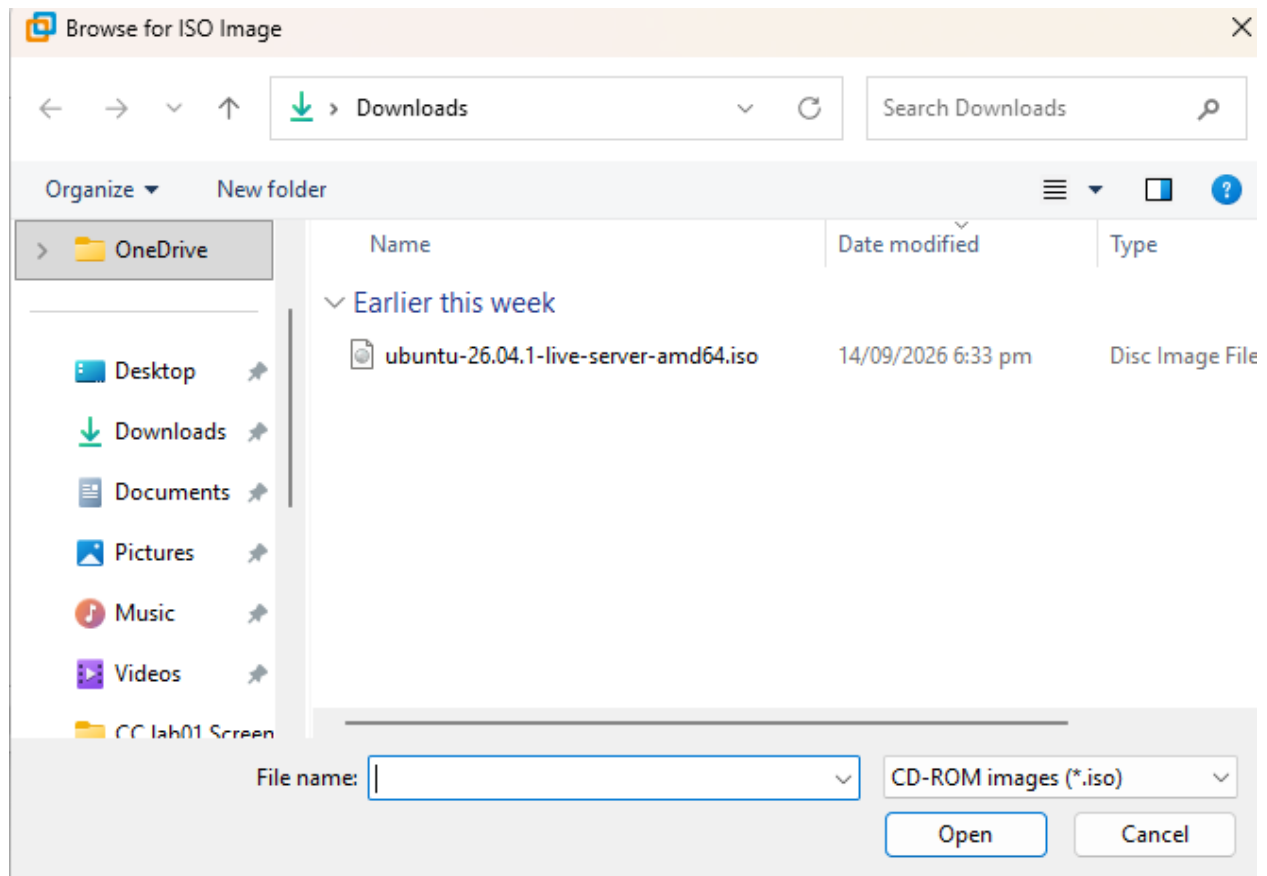
Selected the **Typical (recommended)** configuration option for standard setup parameters.



Selected Typical virtual machine configuration type.

- **Step 3: Select Ubuntu**

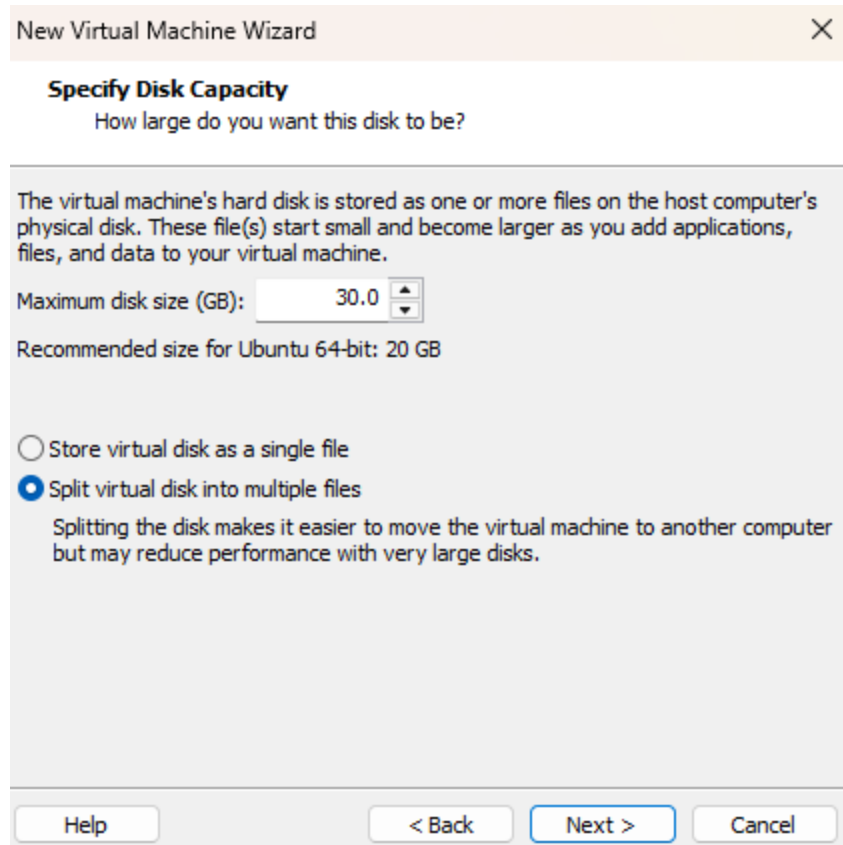




Selected Ubuntu Server ISO file path in VMware setup.

- **Step 4: Allocate Storage & Review Configuration**

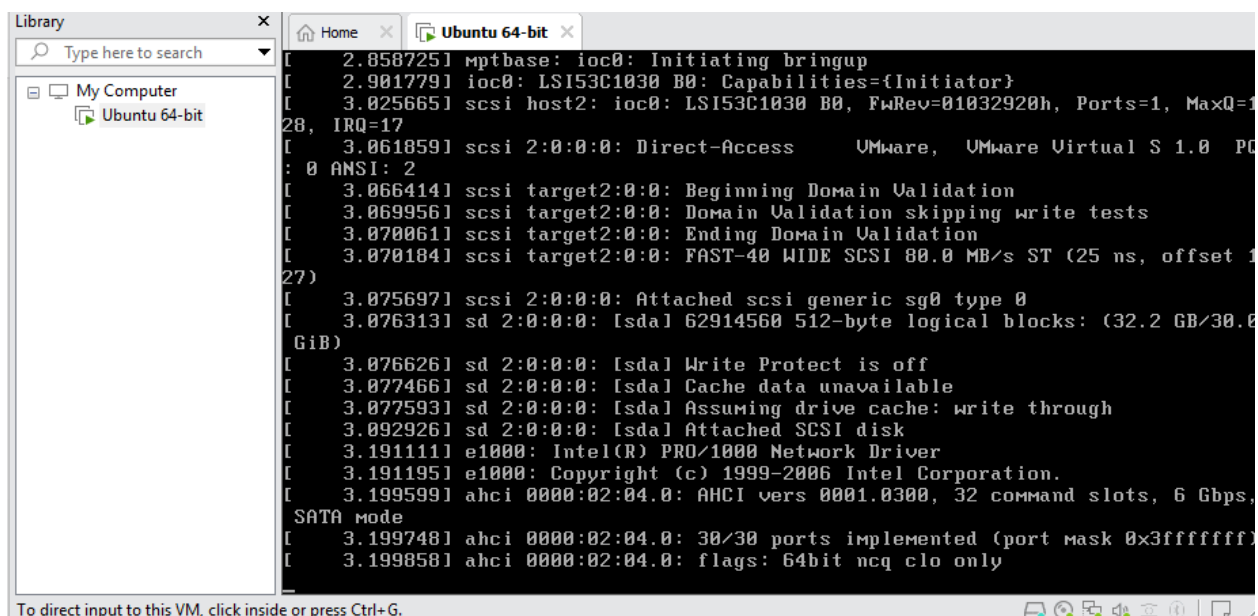
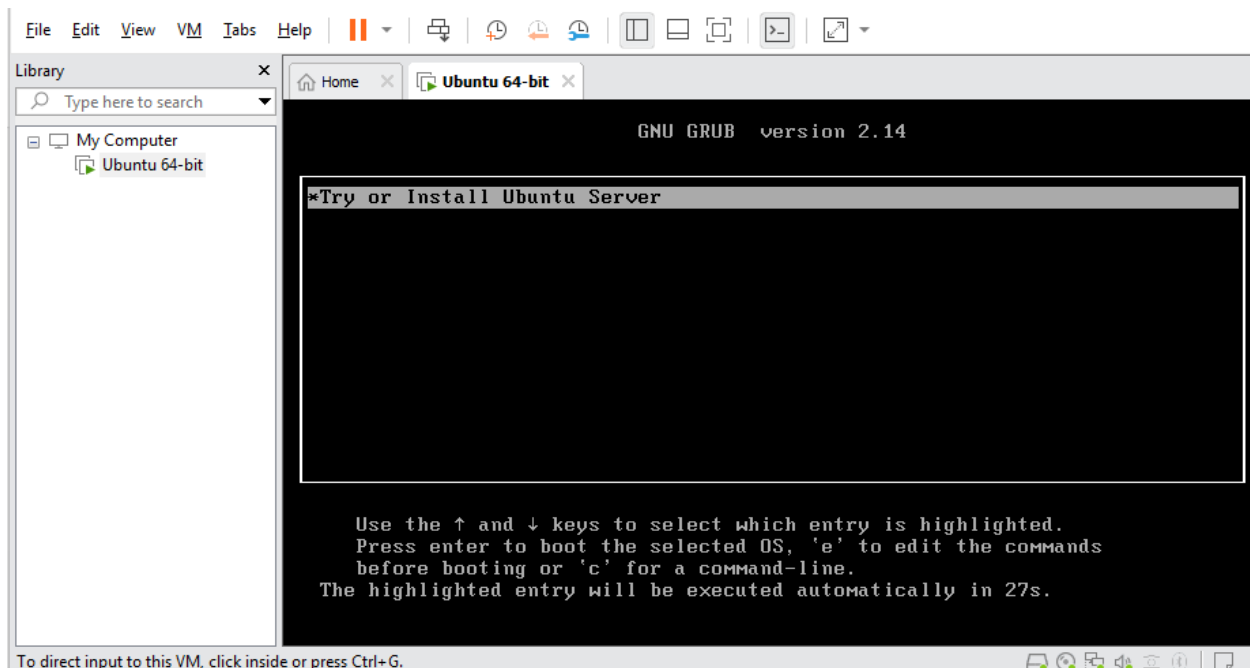
Allocated at least 20 GB of virtual disk space and finalized the virtual machine parameters.



Final virtual machine hardware and storage configuration summary.

- **Step 5: Boot Ubuntu Installer**

Powered on the virtual machine and waited for the installer boot screen to load.



Initial Ubuntu Server installer boot screen in VMware.

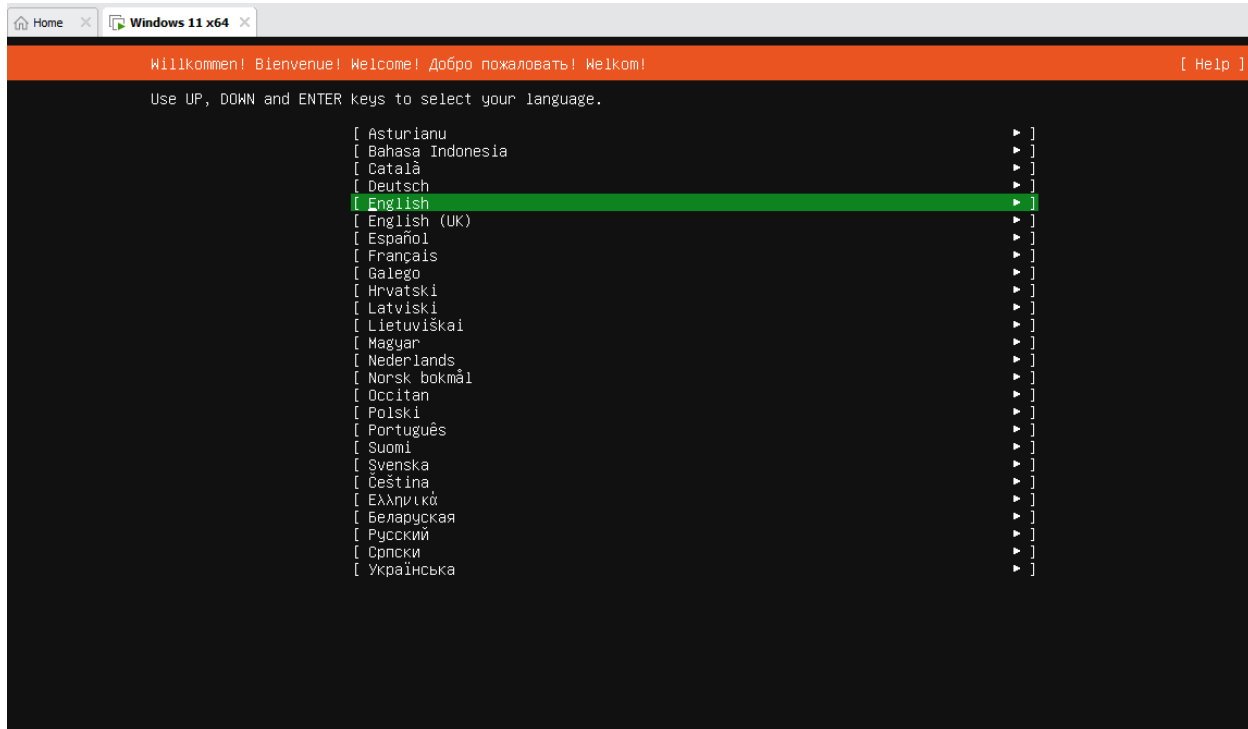
Task 5:

Choose the Installation Language

Selected the preferred language for the installation process.

- **Step 1: Select Language**

Used the arrow keys to choose **English** as the installer language and pressed Enter.



Selected English as the installation language.

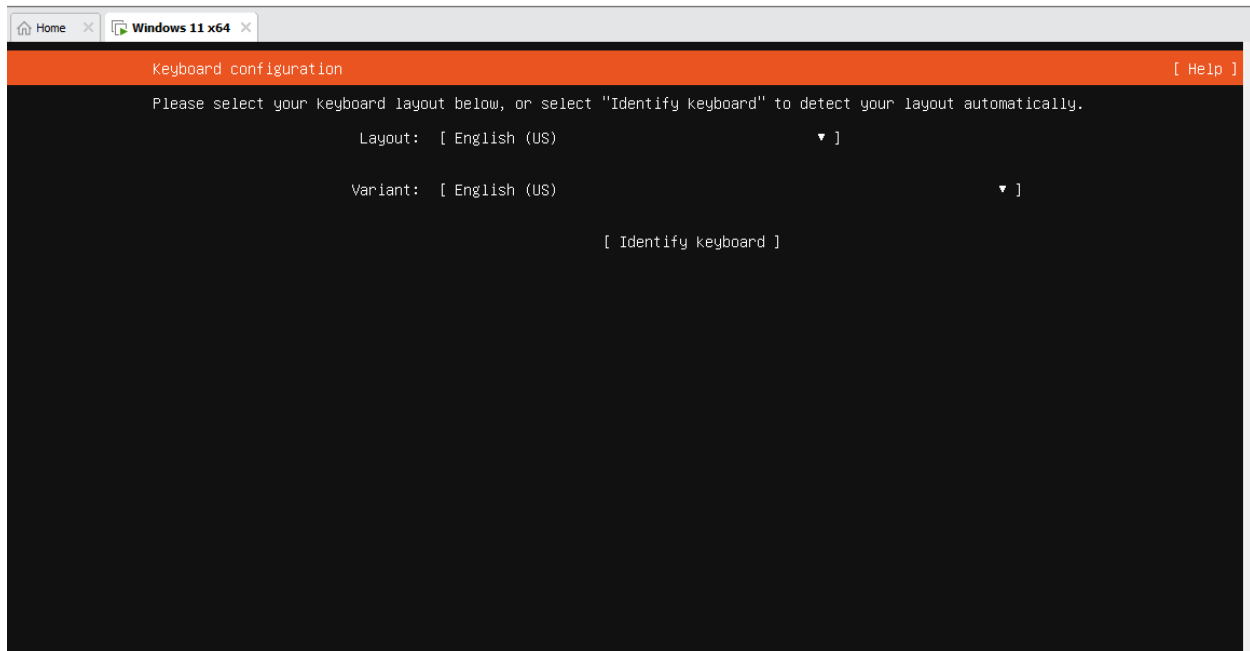
Task 6

Configure the Keyboard Layout

Configured the keyboard input settings matching the hardware configuration.

- **Step 1: Keyboard Layout Selection**

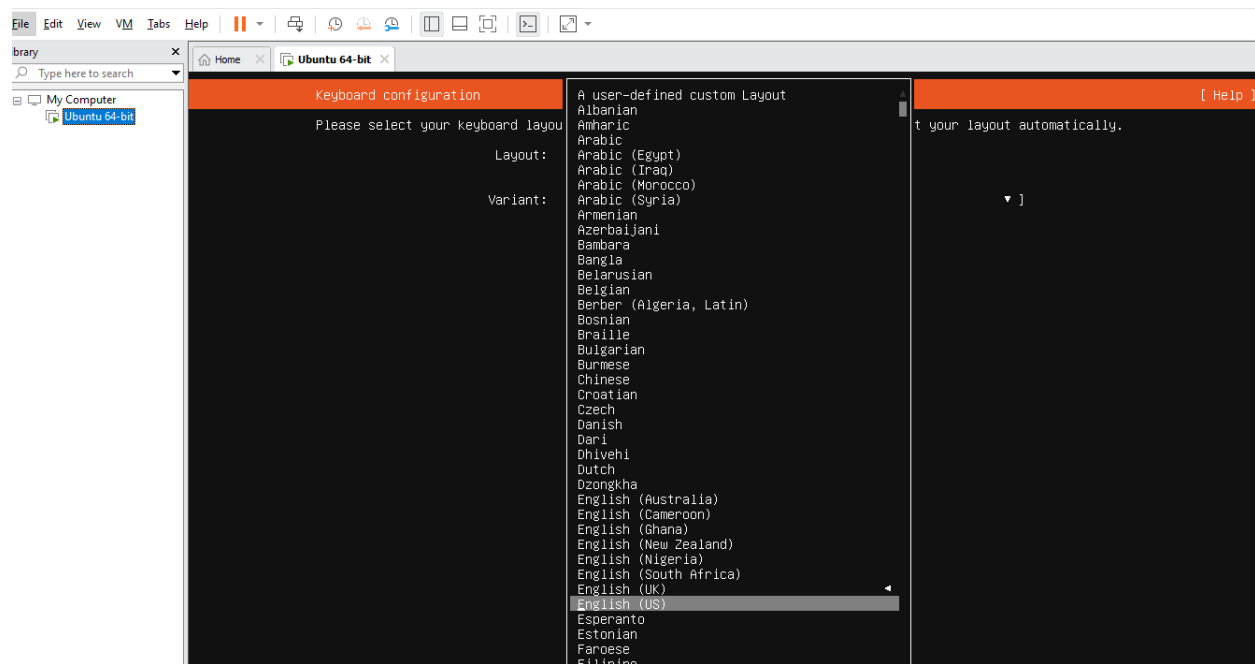
Selected the standard keyboard layout matching the hardware keyboard.

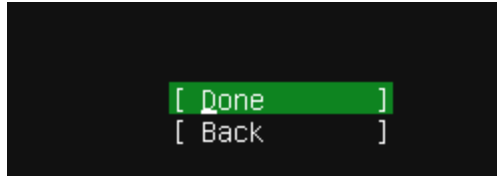


Selected keyboard layout selection screen.

- **Step 2: Keyboard Variant Selection**

Selected the appropriate keyboard variant and proceeded by selecting **Done**.





Selected keyboard variant screen.

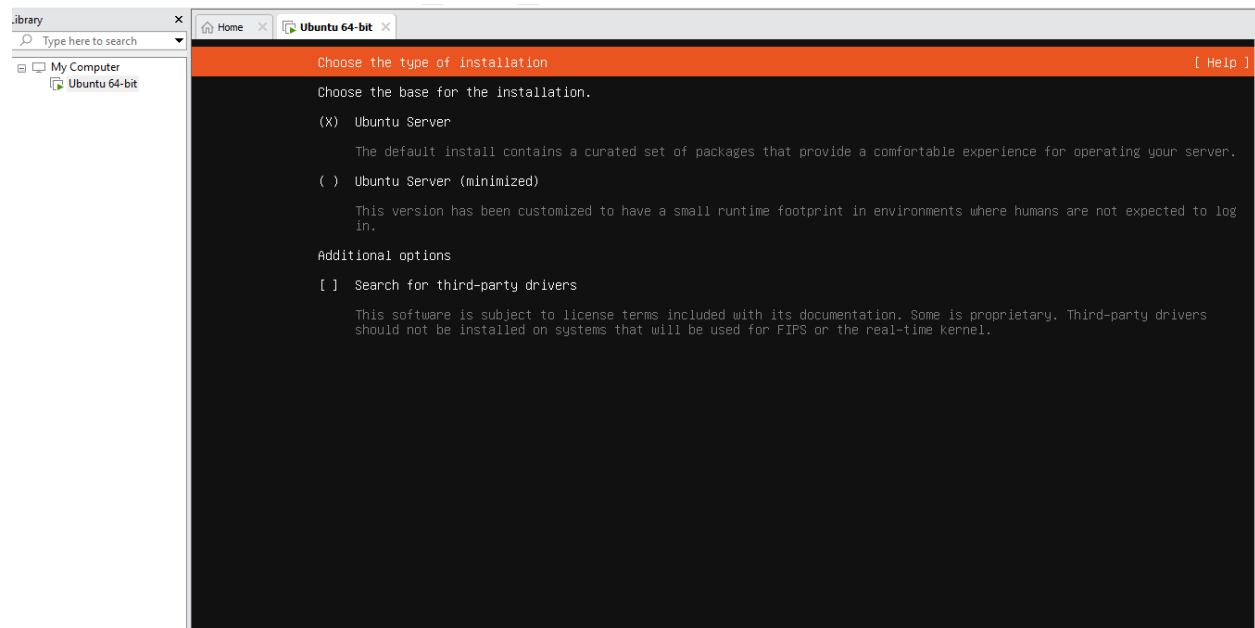
Task 7

Choose the Ubuntu Server Installation Type

Selected the standard installation type for standalone server deployment.

- **Step 1: Installation Type Selection**

Chosen the default **Ubuntu Server** base installation option.



: Selected default Ubuntu Server installation type.

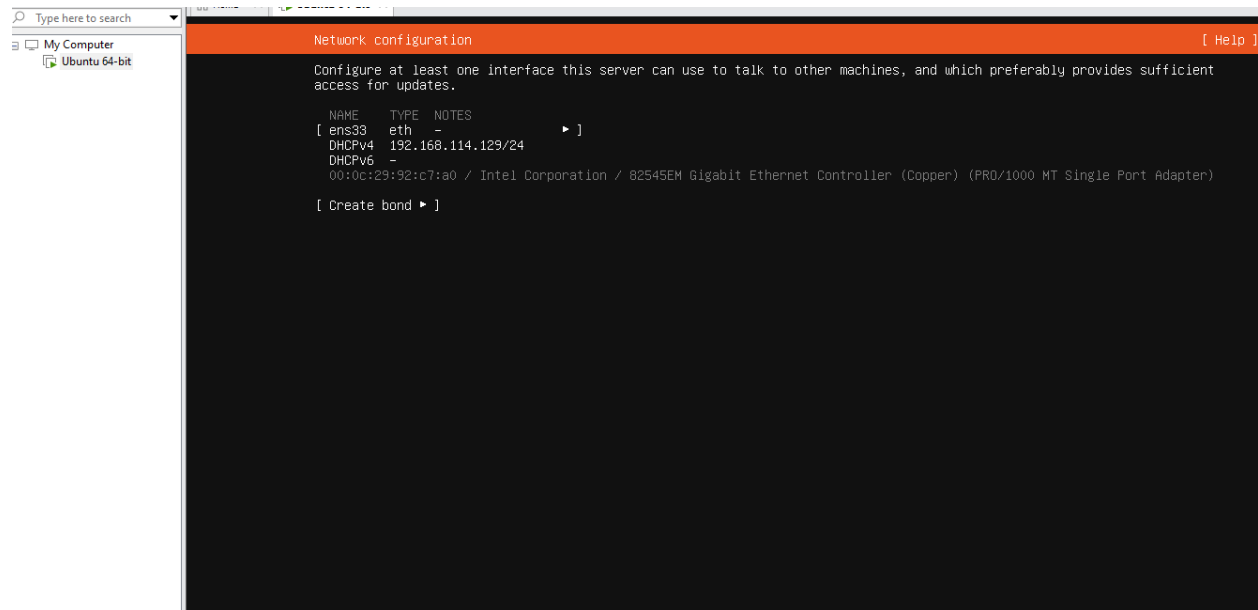
Task 8

Configure the Network

Verified network interface auto-configuration via DHCP.

- **Step 1: Network Interface Detection**

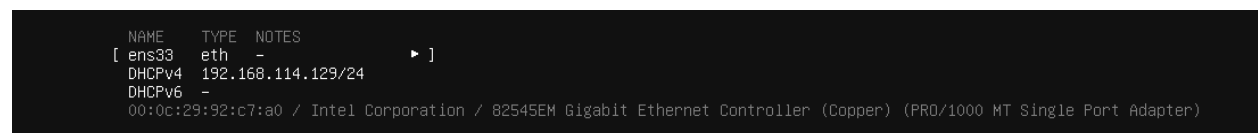
Reviewed the network interface detected by the installer.



Detected network interface in installer.

- **Step 2: DHCP IP Assignment Verification**

Confirmed that the network interface successfully received an IP address dynamically via DHCP.



Dynamic IPv4 address assigned to the server via DHCP.

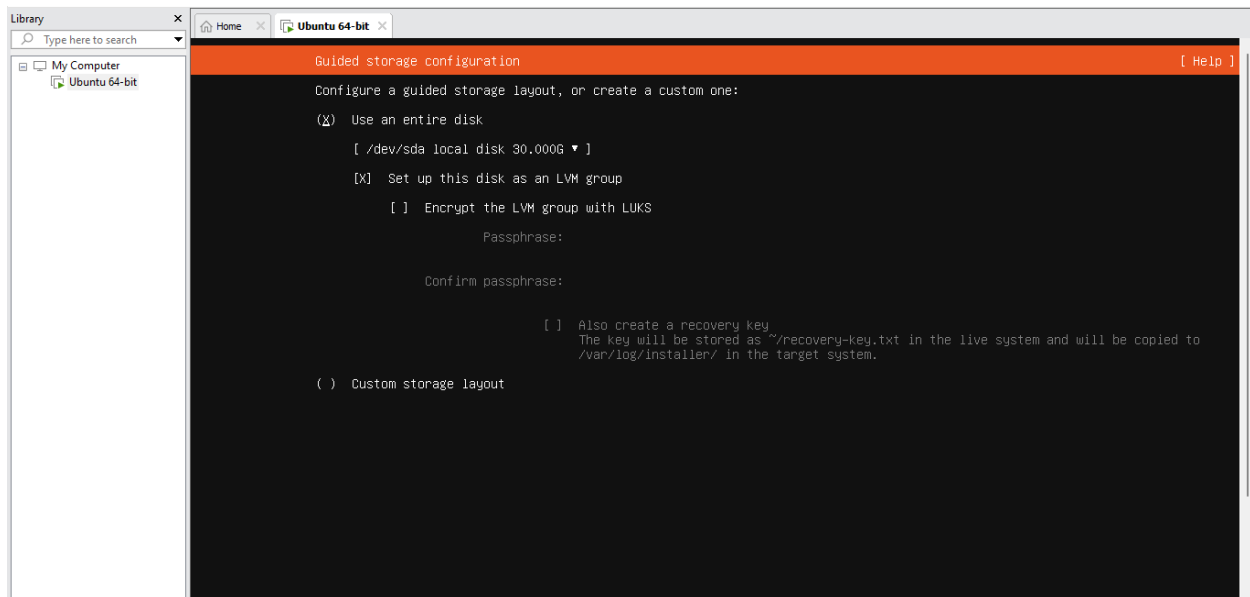
Task 9

Configure Storage

Selected disk partitioning method using the dedicated virtual drive.

- **Step 1: Guided Storage Option**

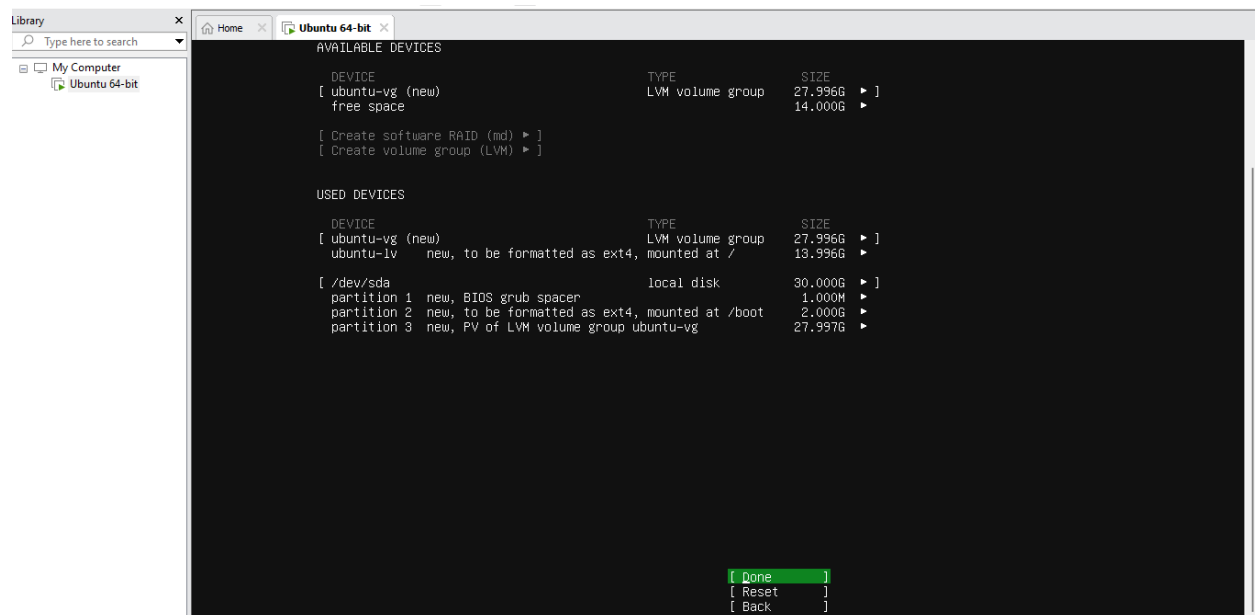
Selected **Use an entire disk** guided storage option targeting the virtual disk.



Selected guided storage option for entire virtual disk.

- **Step 2: Storage Configuration Overview**

Reviewed the automatically configured layout parameters prior to applying changes.



Storage layout summary prior to disk confirmation.

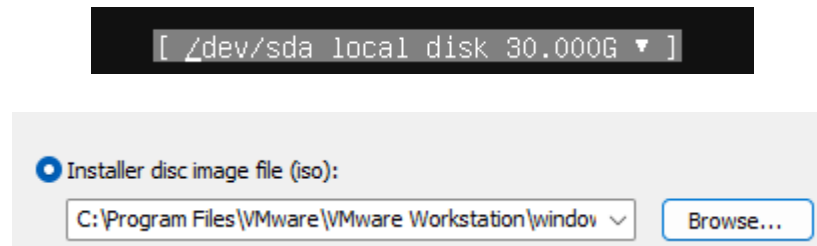
Task 10

Select the Installation Disk

Targeted the exact virtual storage device created in VMware.

- **Step 1: Target Virtual Disk Selection**

Selected the VMware virtual disk for OS deployment.



Selected target VMware virtual disk.

- **Step 2: Disk Details Verification**

Verified disk drive naming, capacity, and volume details.

```
munazzaishfaqalliesshaista056@ubuntu:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                               7:0      0   1.5M  1 loop /snap/hwctl/123
loop1                               7:1      0  66.8M  1 loop /snap/core24/1643
loop2                               7:2      0  50.1M  1 loop /snap/snapd/27710
sda                                 8:0      0   30G   0 disk
├─sda1                             8:1      0    1M   0 part
├─sda2                             8:2      0    2G   0 part /boot
├─sda3                             8:3      0   28G   0 part
│   └─ubuntu--vg-ubuntu--lv 252:0      0   14G   0 lvm  /
sr0                                 11:0     1 1024M   0 rom
```

Displayed virtual disk properties confirming allocated storage capacity.

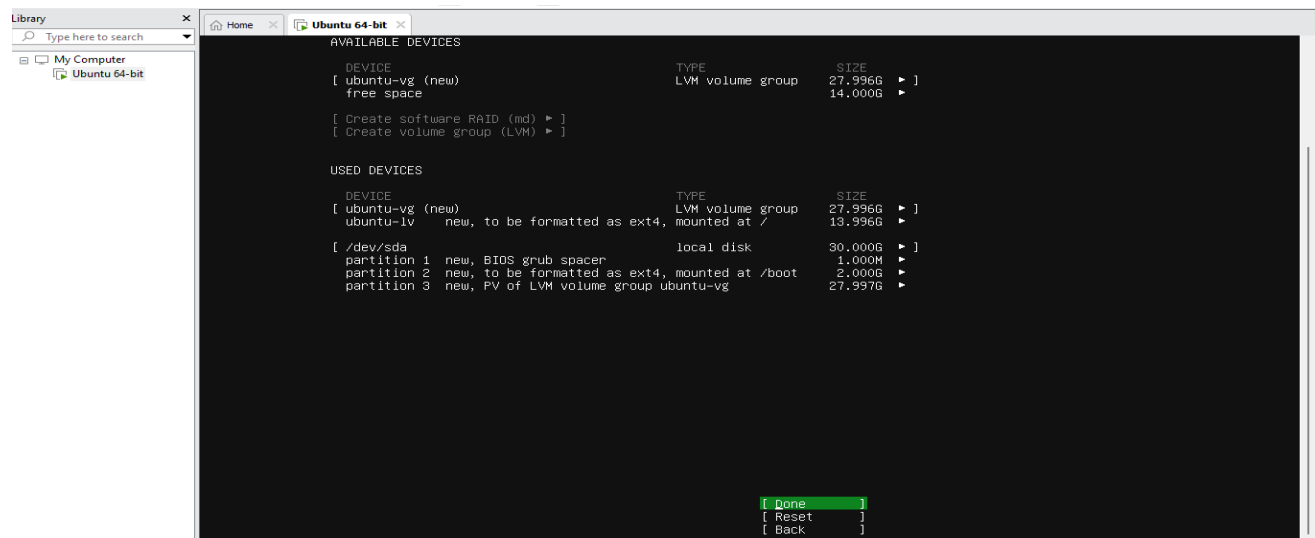
Task 11

Review the Partitions

Analyzed partition geometry generated by the automated installer.

- **Step 1: Partition Scheme Review**

Reviewed the proposed system partitions (root, boot, and LVM volumes).



Automated partition layout scheme on the virtual storage drive.

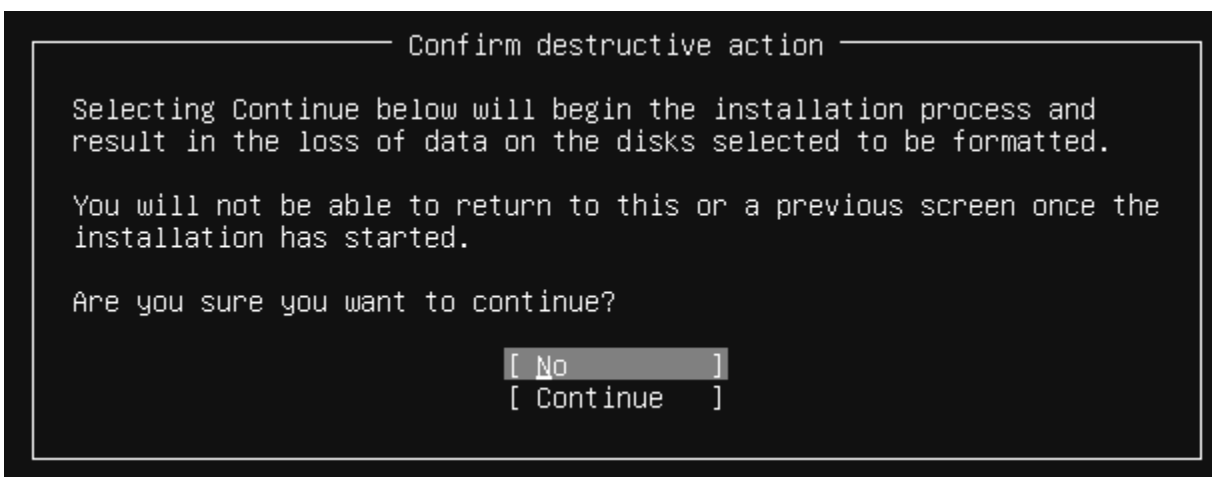
Task 12

Confirm the Storage Changes

Acknowledged destructive partition table writing to the virtual disk.

- **Step 1: Storage Warning**

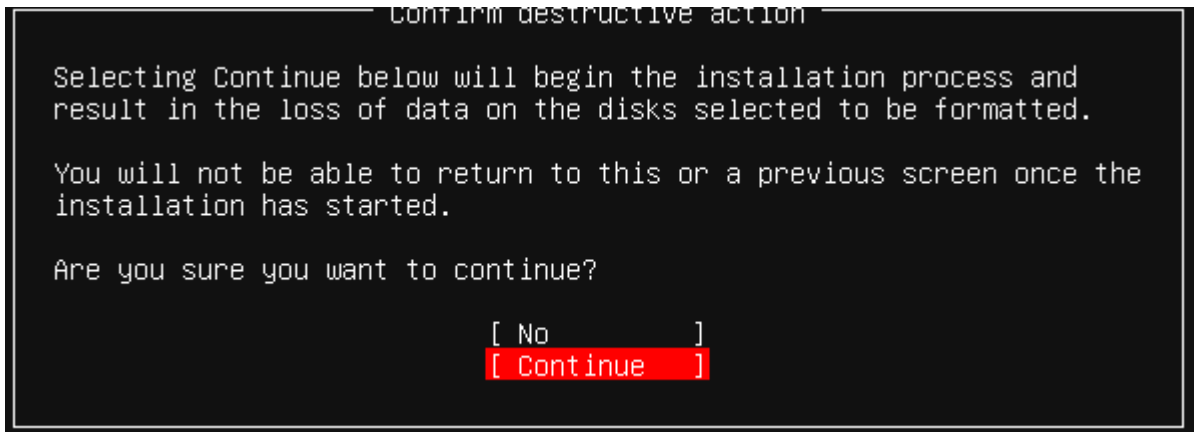
Reviewed the final destructive action warning prompt.



Installer disk write warning message.

- **Step 2: Confirm Partitioning**

Selected **Continue** to execute drive format and partition changes on the target drive.



Final confirmation to overwrite storage and apply partition changes.

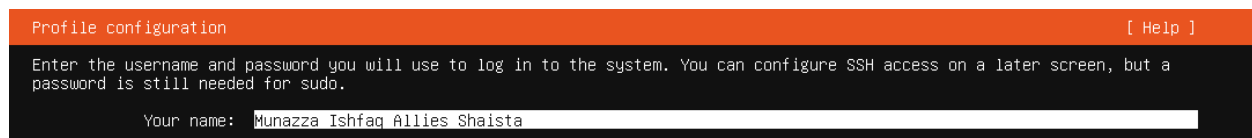
Task 13

Configure the Ubuntu Profile

Configured system hostname, user credentials, and full identity matching lab guidelines.

- **Step 1: Complete Name Entry**

Entered full actual name into the **Your name** profile field.



Completed actual name field in profile setup.

- **Step 2: Hostname Configuration**

Configured **ubuntu** as the exact system server hostname in **Your server's name** field.



Configured 'ubuntu' as the server hostname.

- **Step 3: Username Configuration**

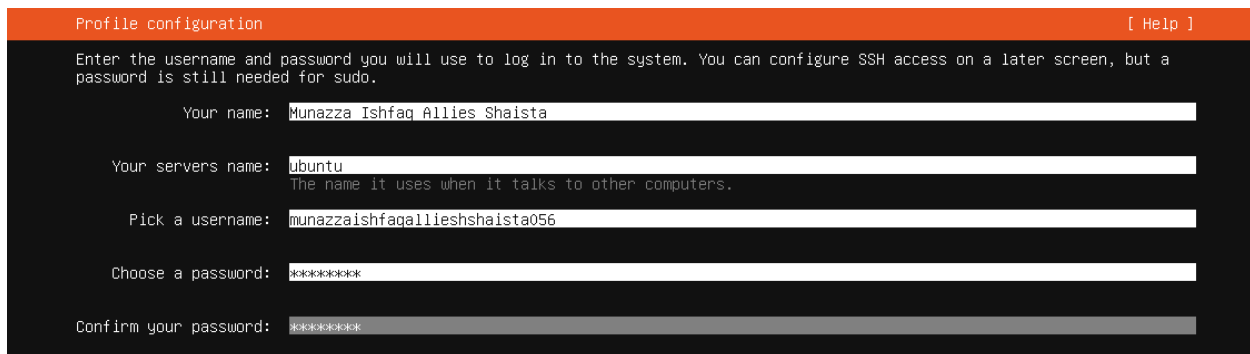
Entered exact GitHub username in lowercase format inside **Pick a username** field.

A screenshot of a terminal window showing the 'Pick a username:' prompt. The text 'munazzaishfaqallieshaista056' is entered into the input field.

Configured system username matching exact GitHub account username.

- **Step 4: Final Profile Review**

Completed user credentials configuration and verified profile fields before continuing.

A screenshot of a terminal window titled 'Profile configuration' with a '[Help]' link in the top right. The screen displays instructions: 'Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.' Below this, several fields are shown with their values: 'Your name: Munazza Ishfaq Allies Shaista', 'Your servers name: ubuntu' (with a sub-note 'The name it uses when it talks to other computers.'), 'Pick a username: munazzaishfaqallieshaista056', 'Choose a password: *****', and 'Confirm your password: *****'.

Completed profile setup page with verified identity parameters.

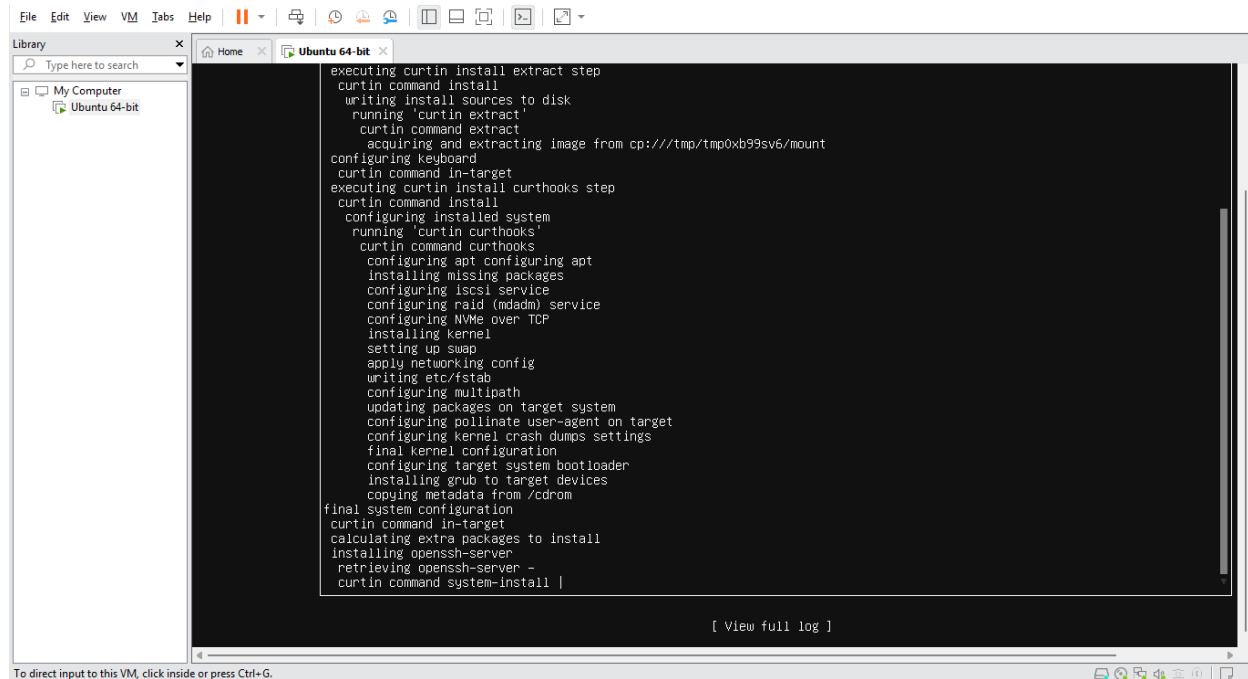
Task 14

Complete the Software Installation

Monitored base OS package extraction and kernel setup execution.

- **Step 14.1: Software Installation Progress**

Observed system package installation progress on the target storage.



Software extraction and base installation progress indicator.

Task 15

Reboot and Verify the Ubuntu Identity

Finalized base setup, restarted server, and logged into terminal interface.

- **Step 1: Installation Completion**

Selected **Reboot Now** option upon completion of package installation.

```
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

```
[ View full log ]
[ Reboot Now   ]
```

Installation complete status notification.

- **Step 2: System Login Screen**

Allowed server to boot and reached the local console login prompt.

```
Ubuntu 26.04.1 LTS ubuntu tty1

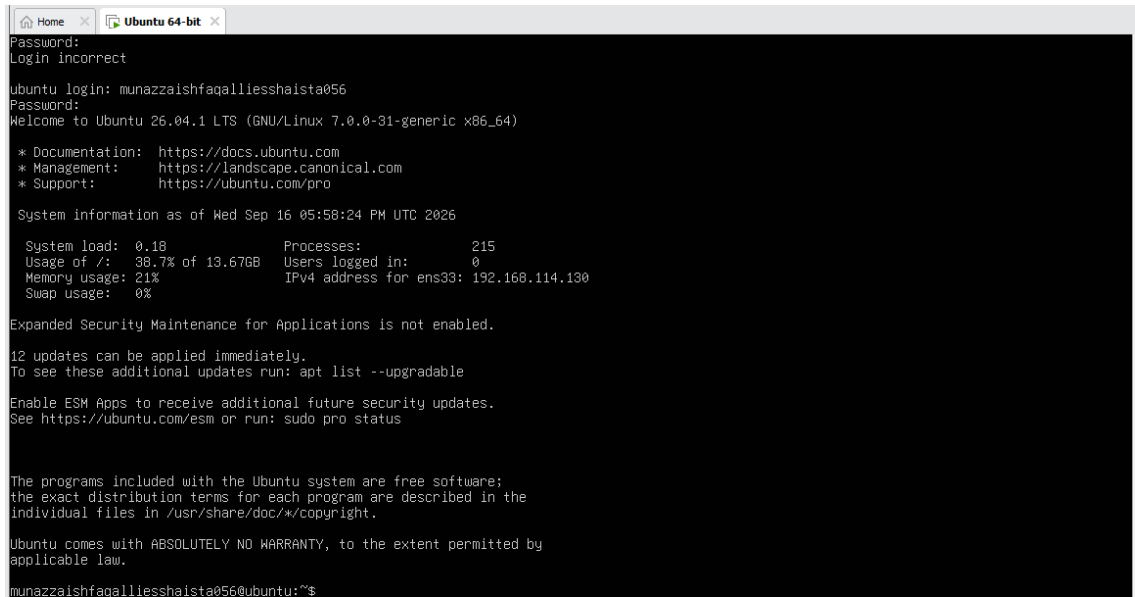
Try contacting this VM's SSH server via 'ssh vsock%2995963808' from host.

ubuntu login: munazzaishfaqalliesshaista056
Password:
```

Ubuntu Server login interface.

- **Step 3: Successful Local Terminal Login**

Logged into the system using the configured credentials.



```
Home x Ubuntu 64-bit x
Password:
Login incorrect
ubuntu login: munazzaishfaqalliesshaista056
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 05:58:24 PM UTC 2026

System load:  0.18           Processes:           215
Usage of /:   38.7% of 13.67GB Users logged in:          0
Memory usage: 21%           IPv4 address for ens33: 192.168.114.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

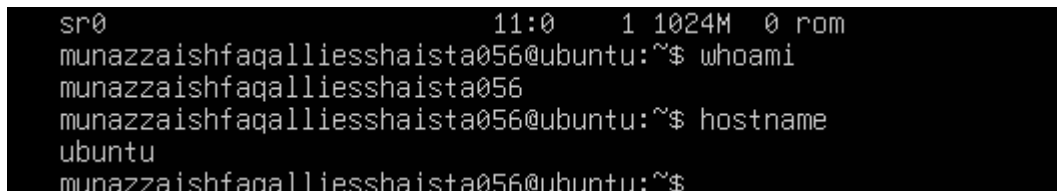
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

munazzaishfaqalliesshaista056@ubuntu:~$
```

Authenticated local terminal login session.

- **Step 4: Terminal Prompt Identity Verification**

Verified system prompt layout matching <github-username>@ubuntu:~\$ syntax requirement.



```
sr0          11:0  1 1024M  0 rom
munazzaishfaqalliesshaista056@ubuntu:~$ whoami
munazzaishfaqalliesshaista056
munazzaishfaqalliesshaista056@ubuntu:~$ hostname
ubuntu
munazzaishfaqalliesshaista056@ubuntu:~$
```

Terminal prompt showing verified identity <github-username>@ubuntu:~\$.

Task 16

Find the Ubuntu Server IP Address

Retrieved system network adapter address using local diagnostic commands.

- **Step 1: Execute IP Address Command**

Executed ip addr inside local terminal shell.

```
munazzaishfagalliesshaista056@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:92:c7:a0 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2992c7a0
    inet 192.168.114.130/24 metric 100 brd 192.168.114.255 scope global dynamic ens33
        valid_lft 1544sec preferred_lft 1544sec
    inet6 fe80::20c:29ff:fe92:c7a0/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
munazzaishfagalliesshaista056@ubuntu:~$
```

Terminal output of ip addr command execution.

- **Step 2: Identify IPv4 Address**

Identified the primary active network interface IPv4 address under inet parameters.

```
valid_lft 1544sec preferred_lft 1544sec
inet6 fe80::20c:29ff:fe92:c7a0/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
```

Selected active network interface IPv4 address displaying shell prompt.

Task 17

Connect to Ubuntu Server from Windows

Established secure remote shell session from Windows terminal to Ubuntu VM.

- **Step 1: Initiate SSH Connection**

Opened Windows Command Prompt and initiated connection string ssh <github-username>@<ubuntu-server-ip>.

```
munazzaishfaqalliesshaista056 X + v
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\trg87>ssh munazzaishfaqalliesshaista056
ssh: Could not resolve hostname munazzaishfaqalliesshaista056: No such host is known.

C:\Users\trg87>ssh munazzaishfaqallisshaista
ssh: Could not resolve hostname munazzaishfaqallisshaista: No such host is known.

C:\Users\trg87>ssh munazzaishfaqalliesshaista056@192.168.114.130
The authenticity of host '192.168.114.130 (192.168.114.130)' can't be established.
ED25519 key fingerprint is SHA256:LPurBFhQxIP+A9mhGI4fao70flsTbxIKJUMsWyYntwA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.114.130' (ED25519) to the list of known hosts.
munazzaishfaqalliesshaista056@192.168.114.130's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 06:48:01 PM UTC 2026

System load:  0.15           Processes:            221
Usage of /:   38.8% of 13.67GB Users logged in:       0
Memory usage: 22%           IPv4 address for ens33: 192.168.114.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Executing SSH connection string in Windows command environment.

- **Step 2: Accept Server Host Key Fingerprint**

Accepted RSA/ECDSA server host key fingerprint notification by entering yes.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.114.130' (ED25519) to the list of known hosts.
munazzaishfaqalliesshaista056@192.168.114.130's password:
```

Accepting target server SSH host key fingerprint.

- **Step 3: Authenticate SSH Session**

Entered user authentication password and established successful shell access.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.114.130' (ED25519) to the list of known hosts.
munazzaishfaqalliesshaista056@192.168.114.130's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed Sep 16 06:48:01 PM UTC 2026

System load:  0.15           Processes:            221
Usage of /:   38.8% of 13.67GB Users logged in:       0
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Authenticated remote SSH shell session opened from Windows host.

- **Step 4: Verify Remote Session Prompt**

Confirmed remote prompt displays matching identity format <github-username>@ubuntu.

```
Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

munazzaishfaqalliesshaista056@ubuntu:~$ |
```

Remote SSH shell prompt displaying verified user host string.

Task 18

Prepare Lab1 Solution PDF

1. Solution Title Page:

Name: Munazza Ishfaq Allies Shaista

Reg No: 2024-BSE-056

Course Code: Cloud Computing

Section: BSE-5B

GitHub Username: munazzaishfaqalliesshaista056

2. Screenshot of Pdf:

